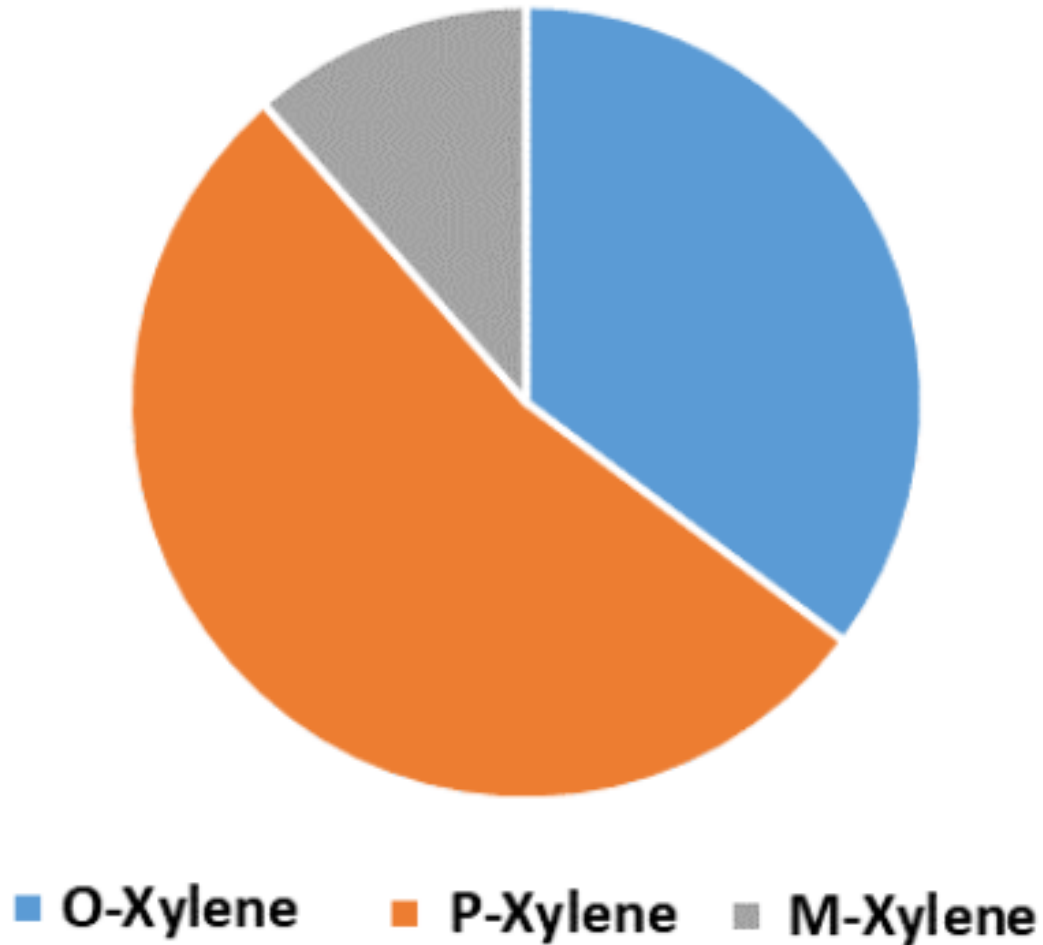


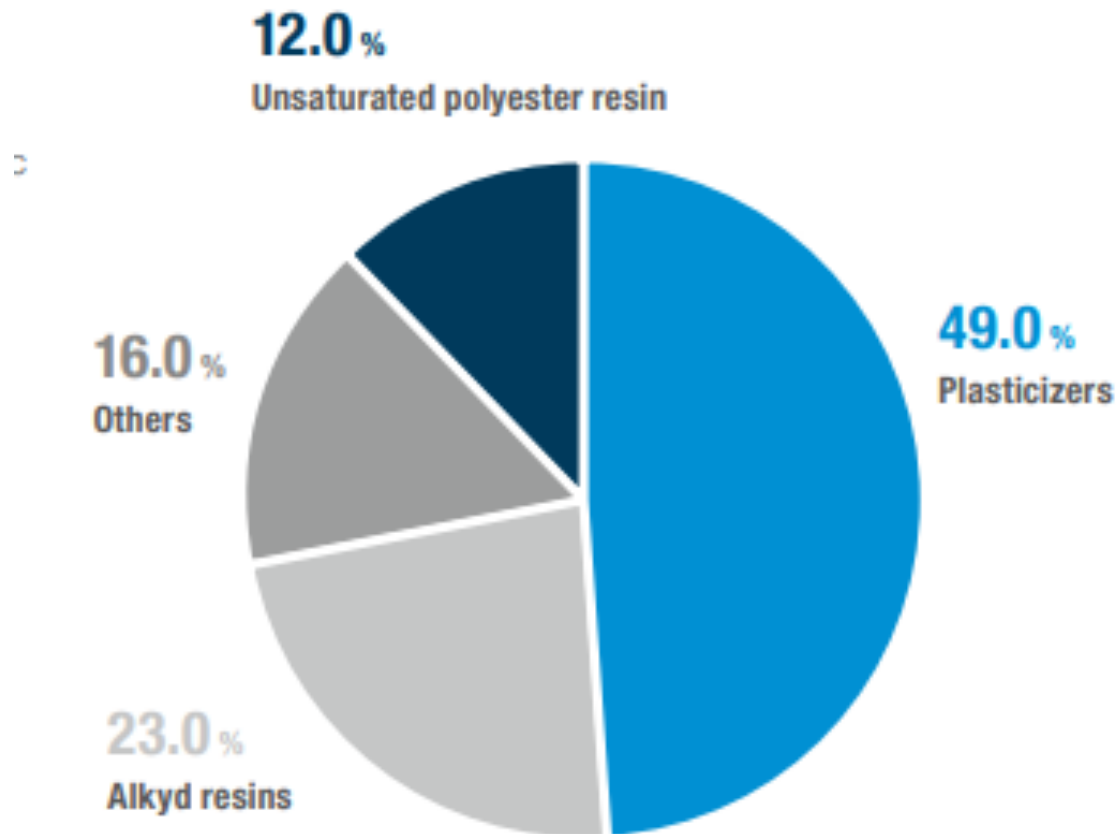
Petrochemical Technology

Chemicals from Xylene

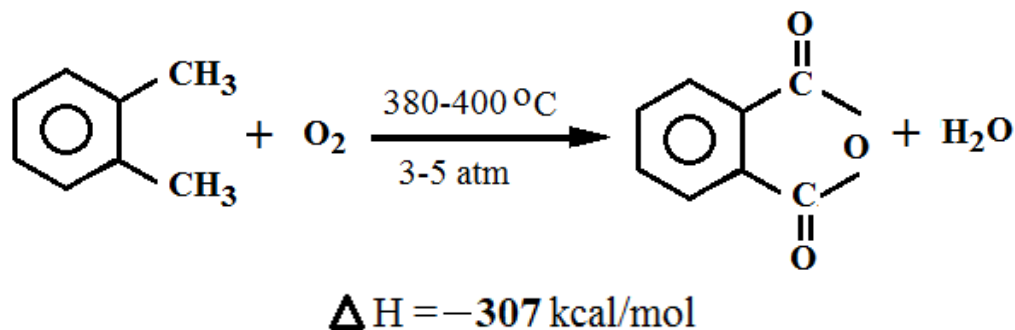
Global Xylene Market, by Type (%)



World Consumption of Phthalic Anhydride by End Use

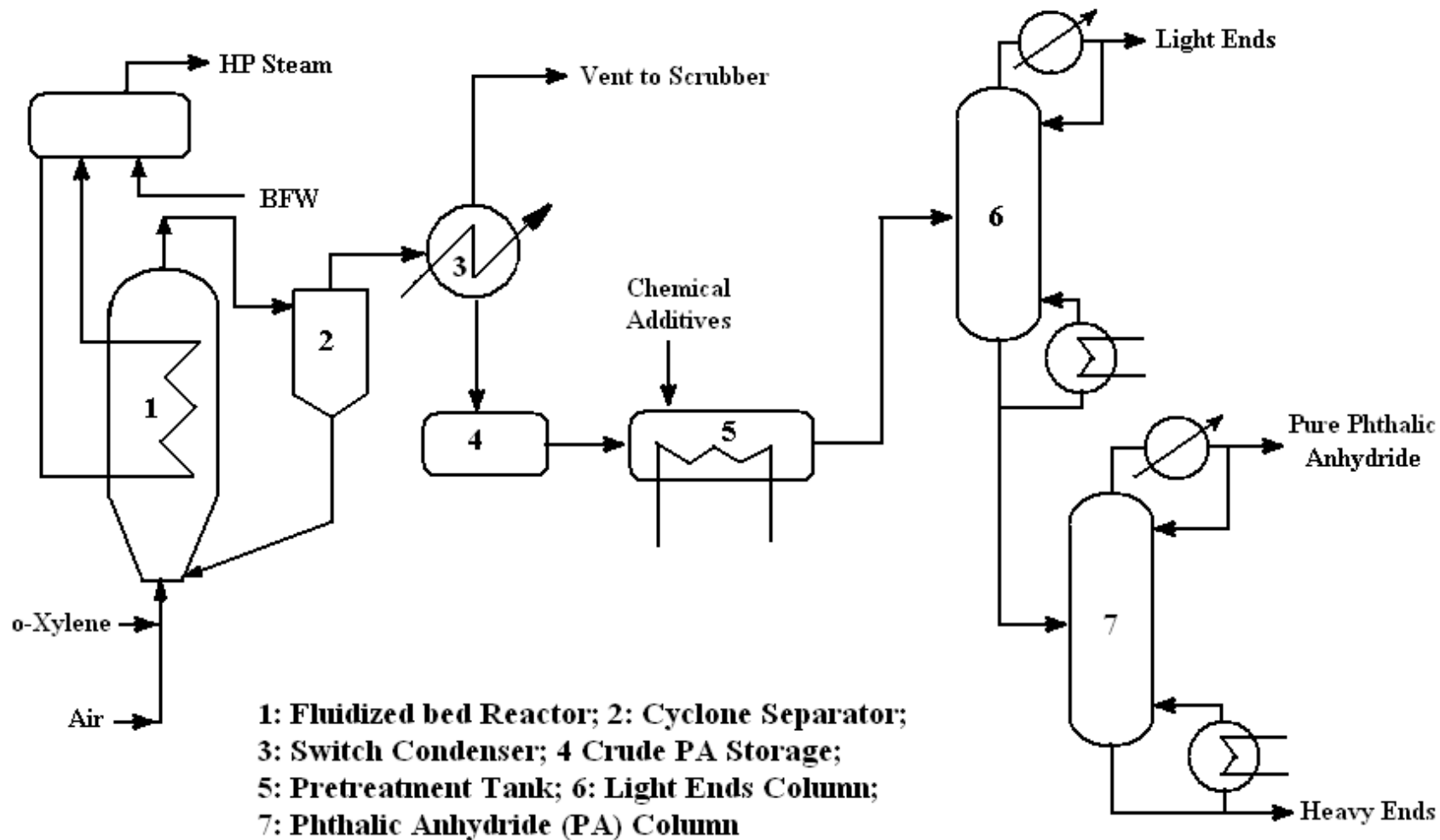


Phthalic Anhydride from *o*-Xylene



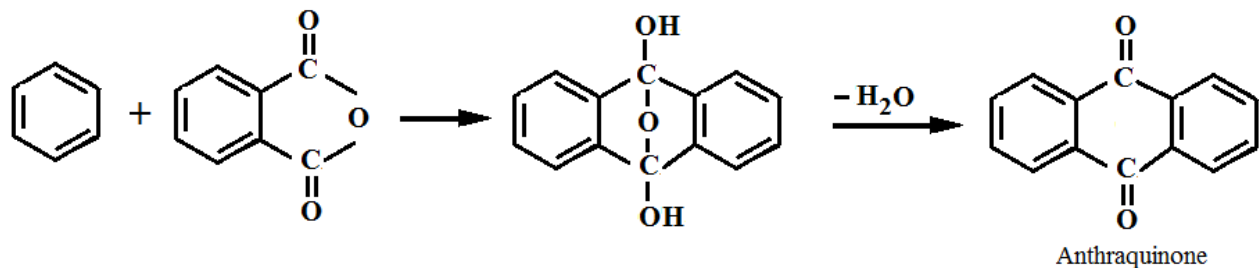
Catalyst: V₂O₅ combined with TiO₂ and these active materials are coated on an inert nonporous carrier. TiO₂ increases the catalyst's affinity for oxygen and improves the selectivity of the reaction to form PA. The yield of PA can be as high as 1.395 kg/kg *o*-X.

The selectivity of the PA reaction is 70%, for the complete combustion of *o*-X is 15%, for the incomplete combustion of *o*-X is 5%, for maleic anhydride is 9% and for the heavy impurity is 1%.



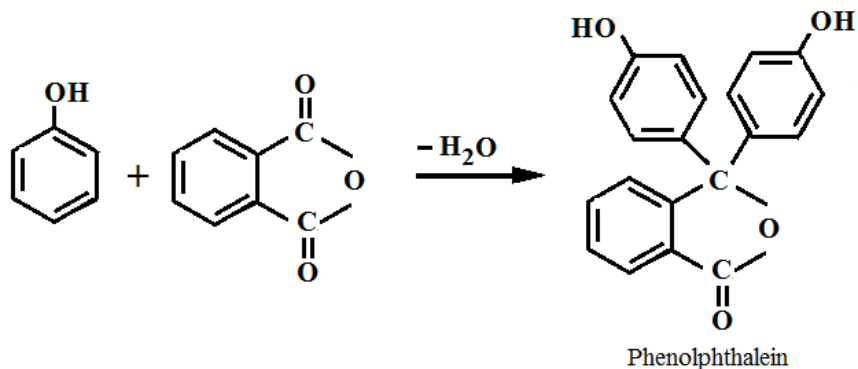
Production of Phthalic Anhydride from *o*-Xylene

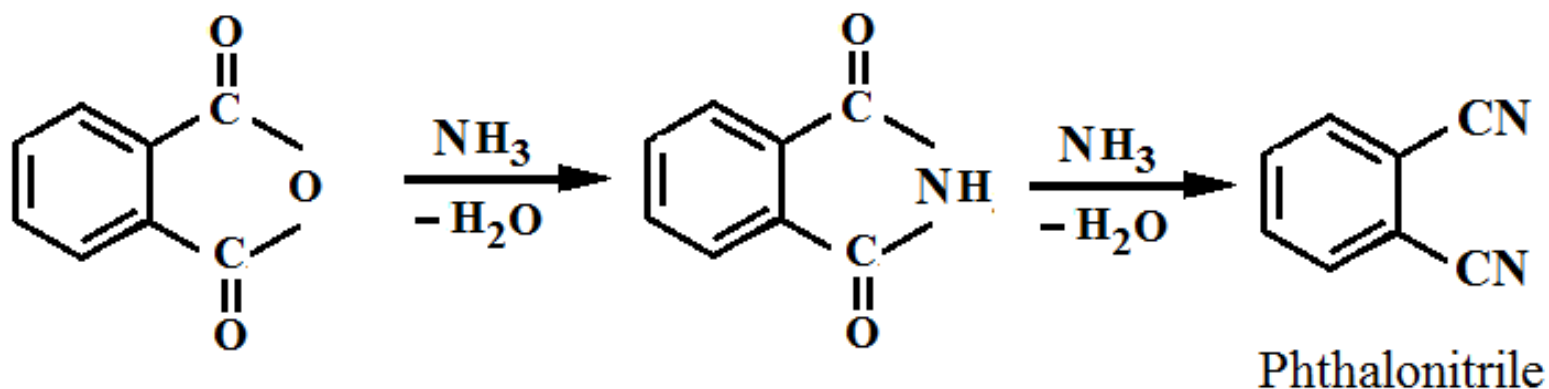
About one half of the phthalic anhydride production is used in plasticizers. One quarter goes into alkyd resins and an equal amount into unsaturated polyesters.



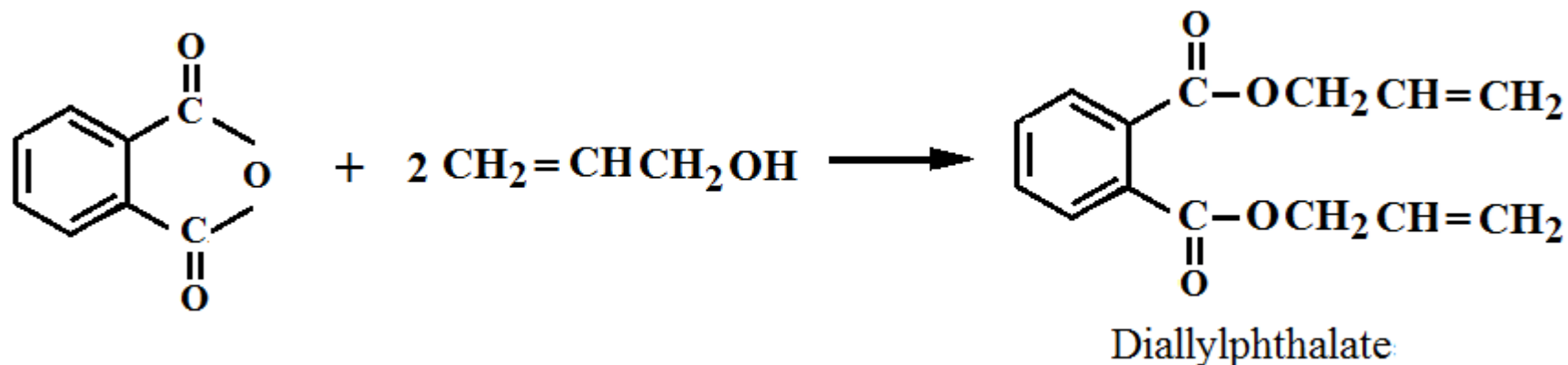
Anthraquinone is used to make dyes, digester additive in paper making and anthracenedione class of drugs.

Phthalic anhydride reacts with phenol in presence of sulphuric acid to form the pH indicator, Phenolphthalein.



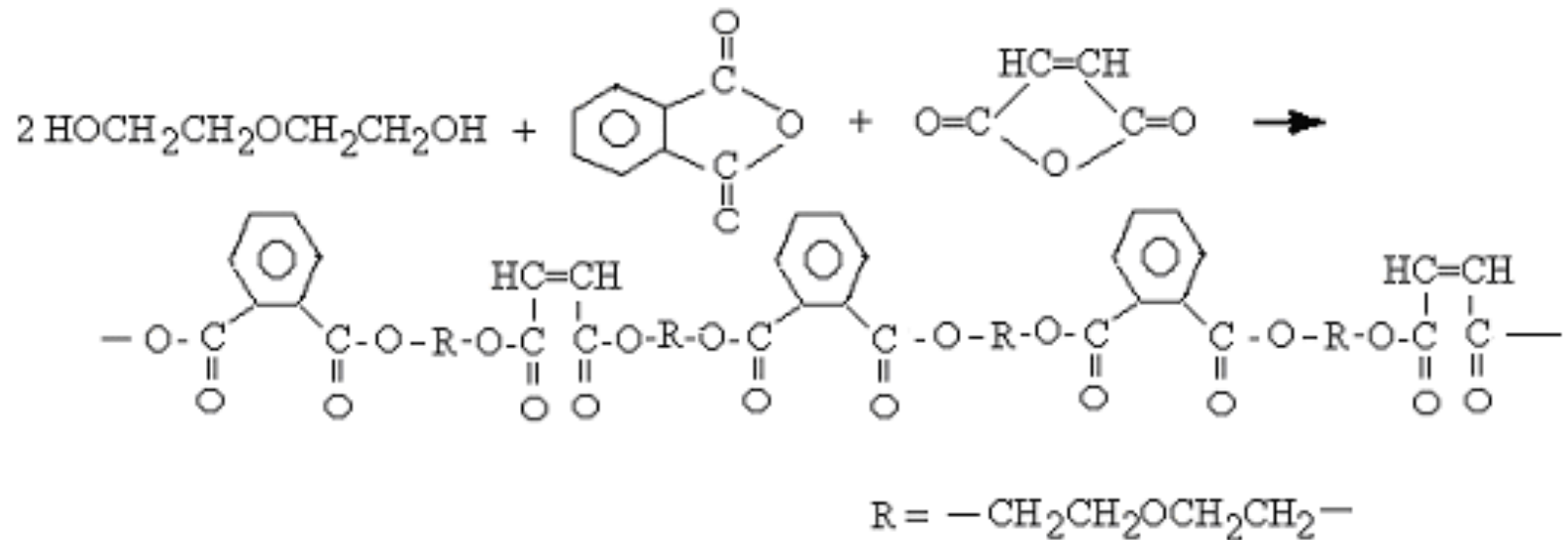


Phthalonitrile is the starting material for phthalocyanine dyes.



Diallylphthalate is used as a monomer for high performance thermoset polymers, useful for glass-reinforced plastics.

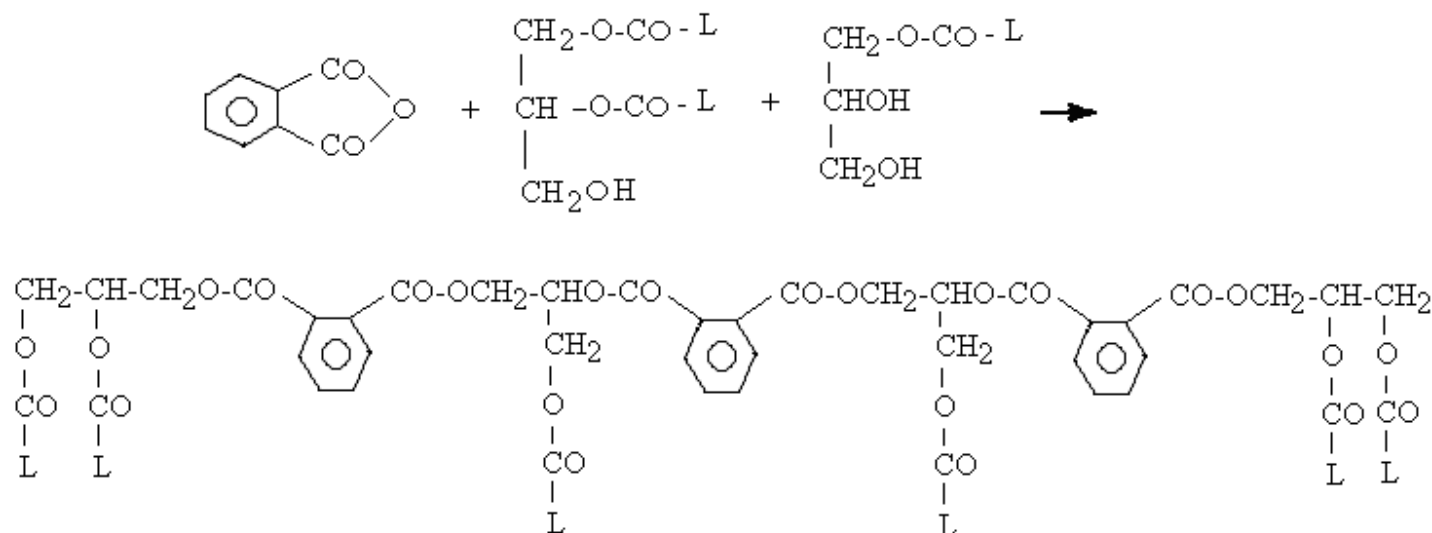
Unsaturated Polyester Resin



Unsaturated polyesters reinforced with glass, cloth or fibers are used for the automobiles. They are used as a metal replacement for the fabrication of pipe and storage tanks, and in the manufacture of small boats and even large minesweepers, where the absence of metal is essential.

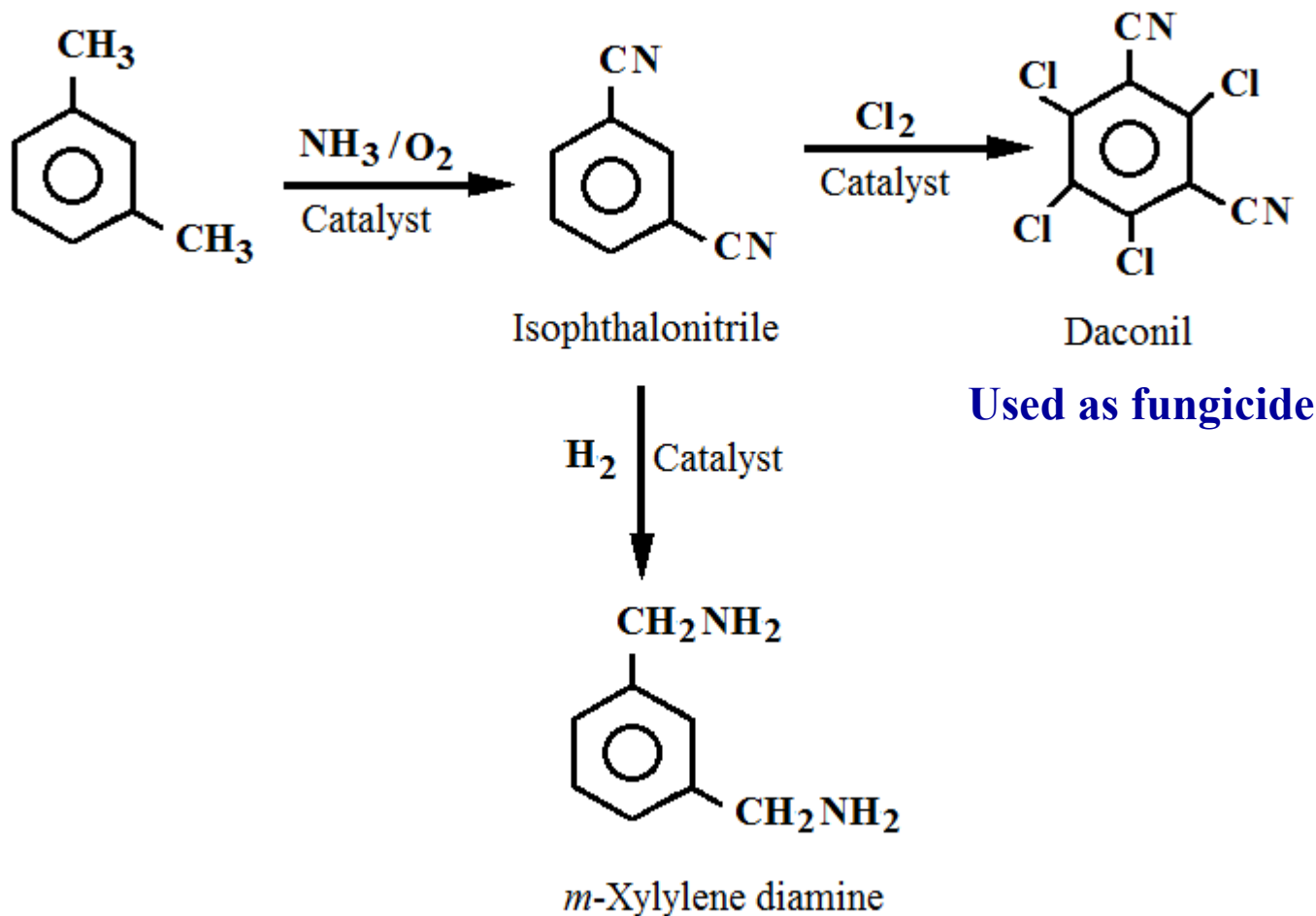
Alkyd Resins

Alkyd resins are oligomers in which polyester functions have been inserted into natural “drying oils”.

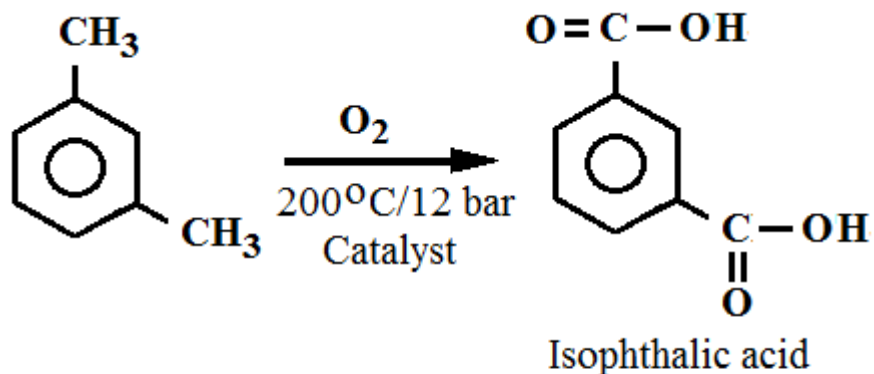


The incorporation of the polyester function gives the paint film greater solvent and ultraviolet resistance and also imparts somewhat greater corrosion-resistance.

Chemicals from *m*-Xylene

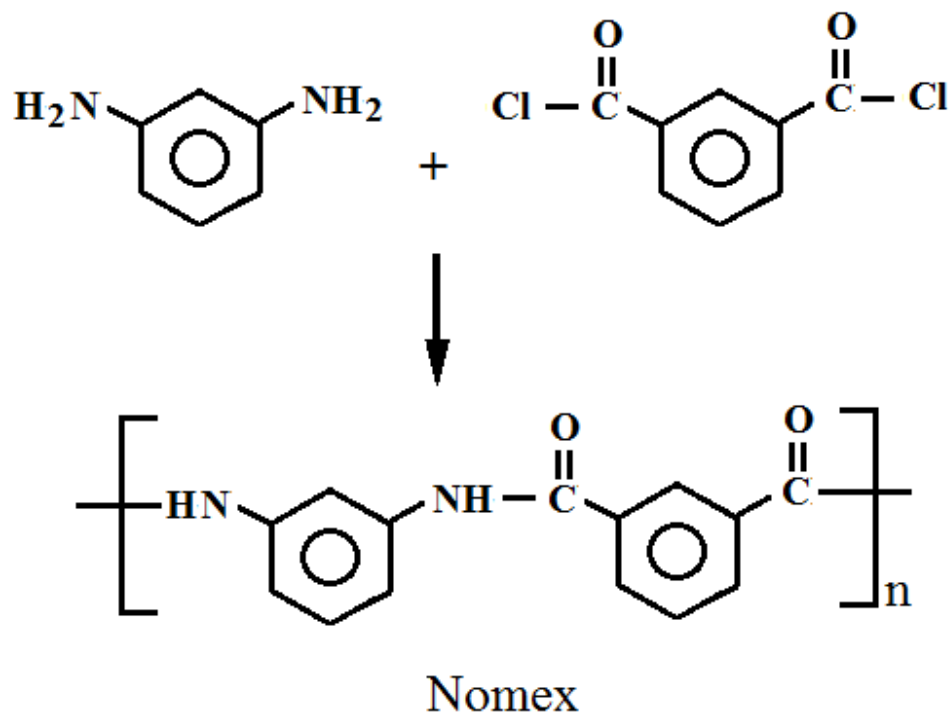


Used as a curing agent
for epoxy resins

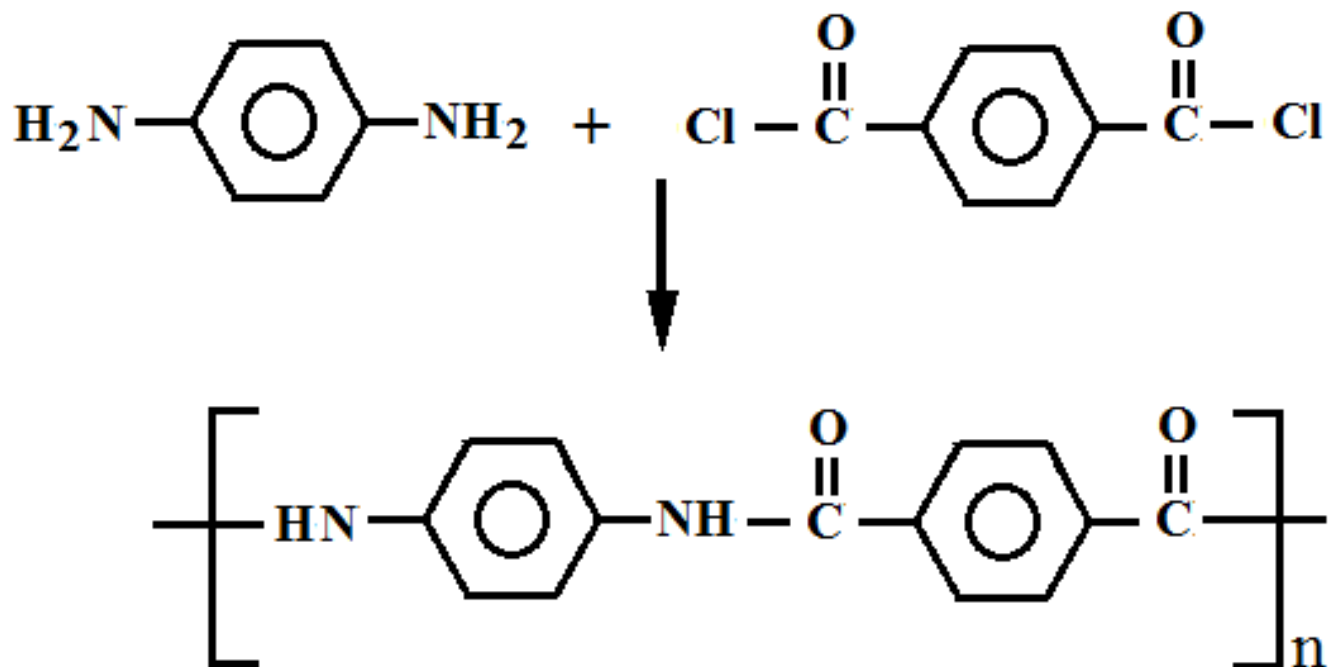


Catalyst: Mn or Co salt with Br as promoter
Or Manganous and Cobaltous
 Bromide in acetic acid medium

**Nomex is used for making
 fire-resistant cloth for fire
 fighters and race car drivers
 uniform and for plant
 workers clothings.**

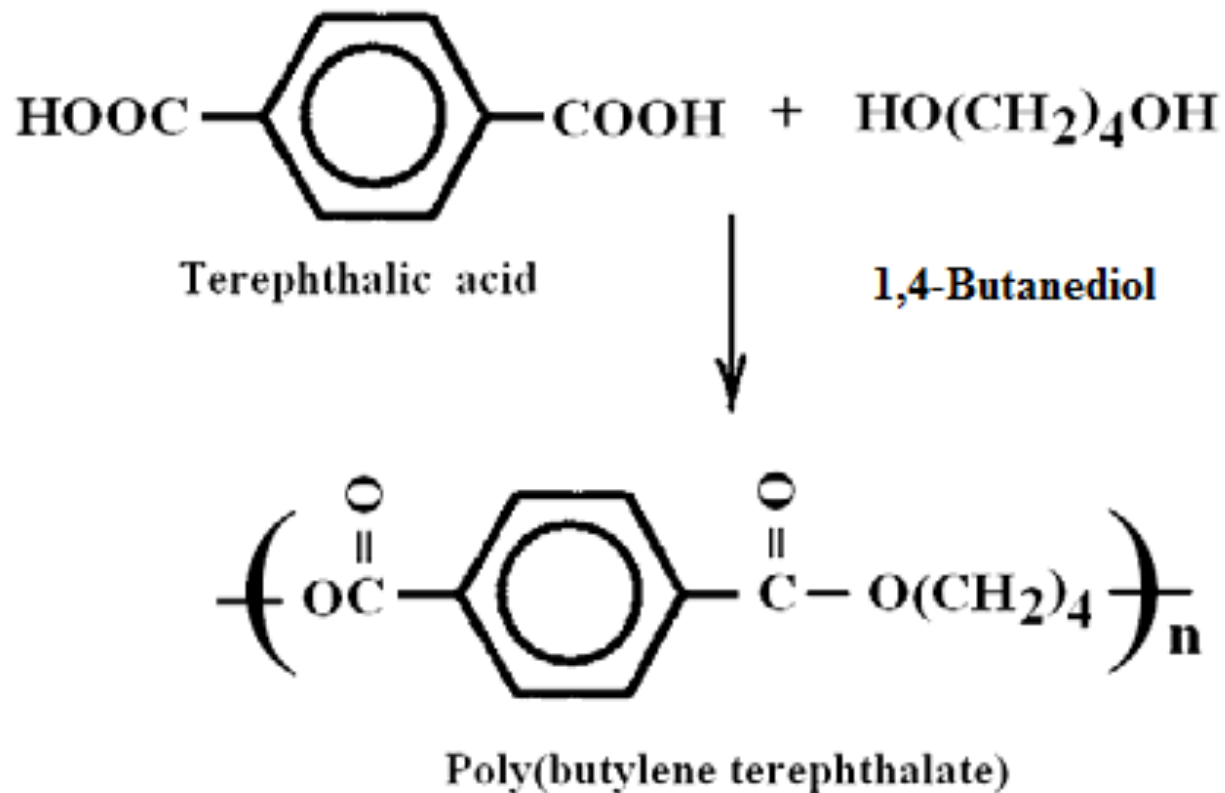


Chemicals from *p*-Xylene

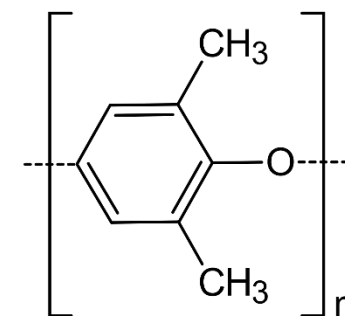


Kevlar

Kevlar finds its major application as an asbestos replacement. Other uses include cord for tires for large vehicles and **reinforcement for bullet-proof vests and military helmets**. It is used in combination with epoxy resin for canoe manufacture and as cloth for boat sails.



PBT is an inexpensive moulding resin useful for pipe, automotive parts, and toothbrush bristles. It is also alloyed with poly(phenylene oxide) to give a plastic useful for automobile bumpers and even for side panels.



PPO

Terephthalic Acid (TPA)

The global purified terephthalic acid (PTA) market is expected to be valued at US\$ 61.78 Billion in 2023. The increasing use of polyester fibres due to their ability to combine with various natural and synthetic fibres is likely to boost demand throughout this decade. The overall demand for purified terephthalic acid (PTA) projected to grow at a CAGR of 5.5% between 2022 and 2032, totaling around US\$ 99.50 Billion by 2032.

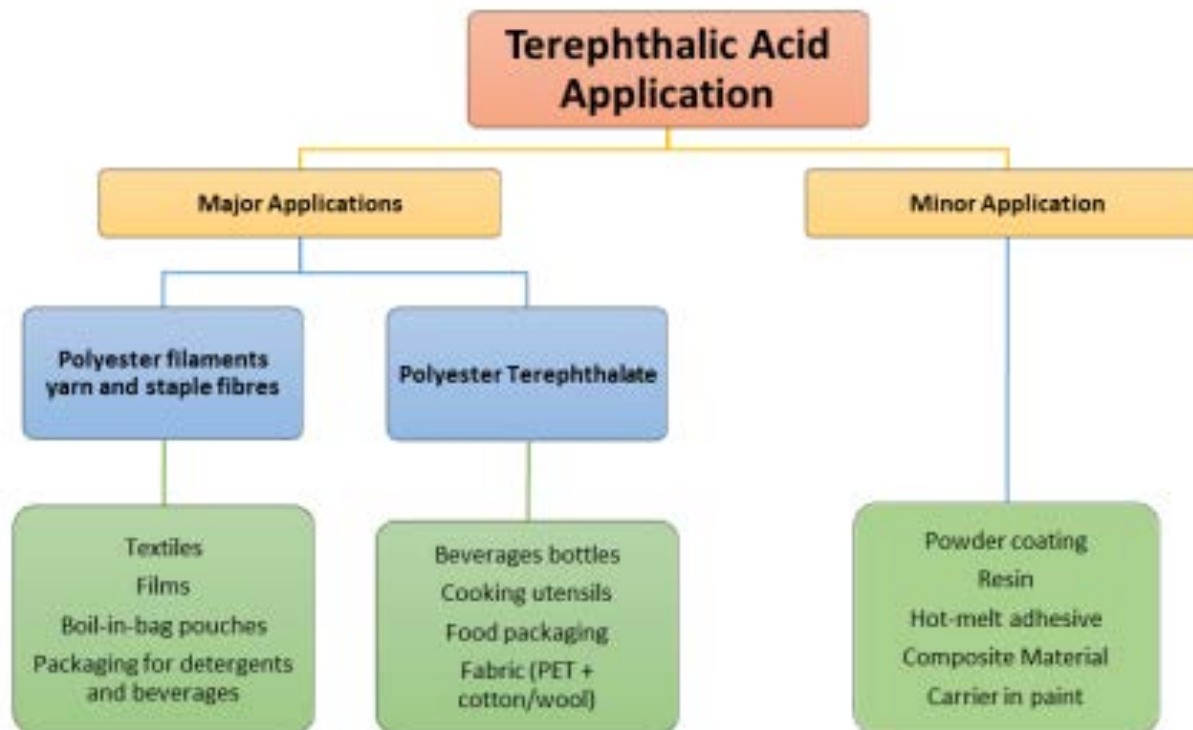
Globally, there is an installed capacity of ca. 81.6 MMT per year, and this is estimated to increase to 105.6 MMT per year by the end of 2029.

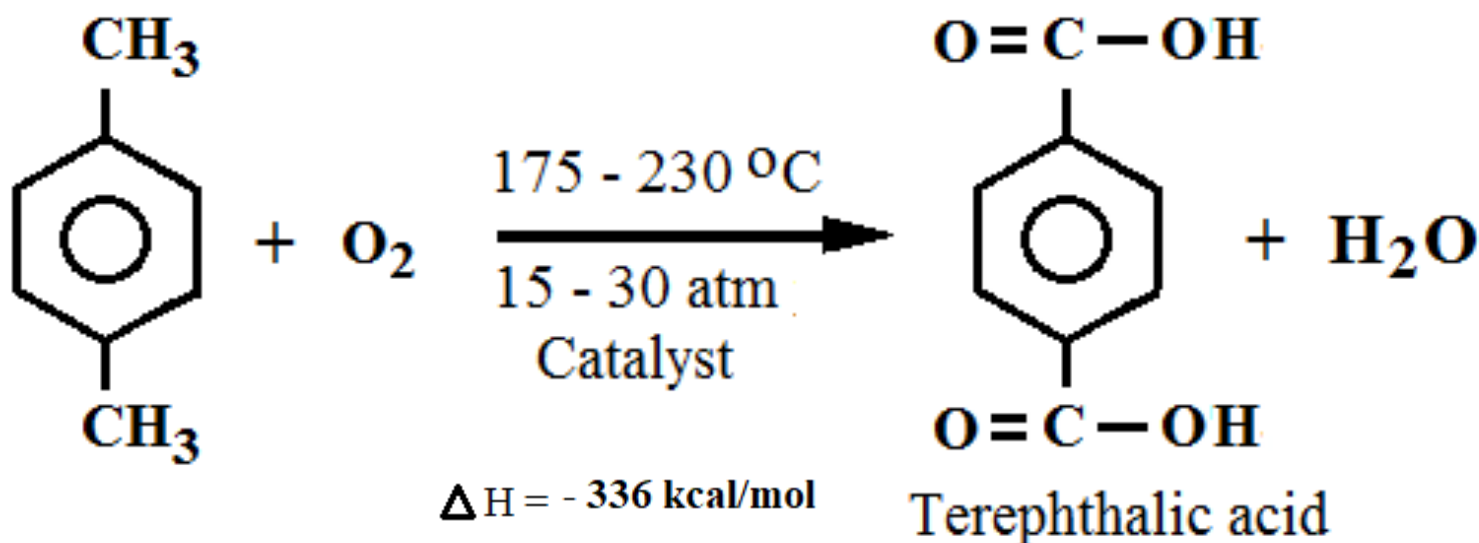
Over the forecast period of 2022-2032, the worldwide purified terephthalic acid market is expected to develop at a high CAGR. Market participants are investigating development opportunities in the maritime, construction, transportation, and wind energy industries.

Applications

Purified terephthalic acid (PTA) is a critical component in the production of synthetic fibres. Purified terephthalic acid accounts for 70-80% of polyester goods. PTA is a less expensive alternative to dimethyl terephthalate (DMT). PTA is widely used in the textile and packaging industries.

Furthermore, PTA is being used in the medical field to treat peptic ulcers, reflux esophagitis, and gastroesophageal reflux disease (GERD).

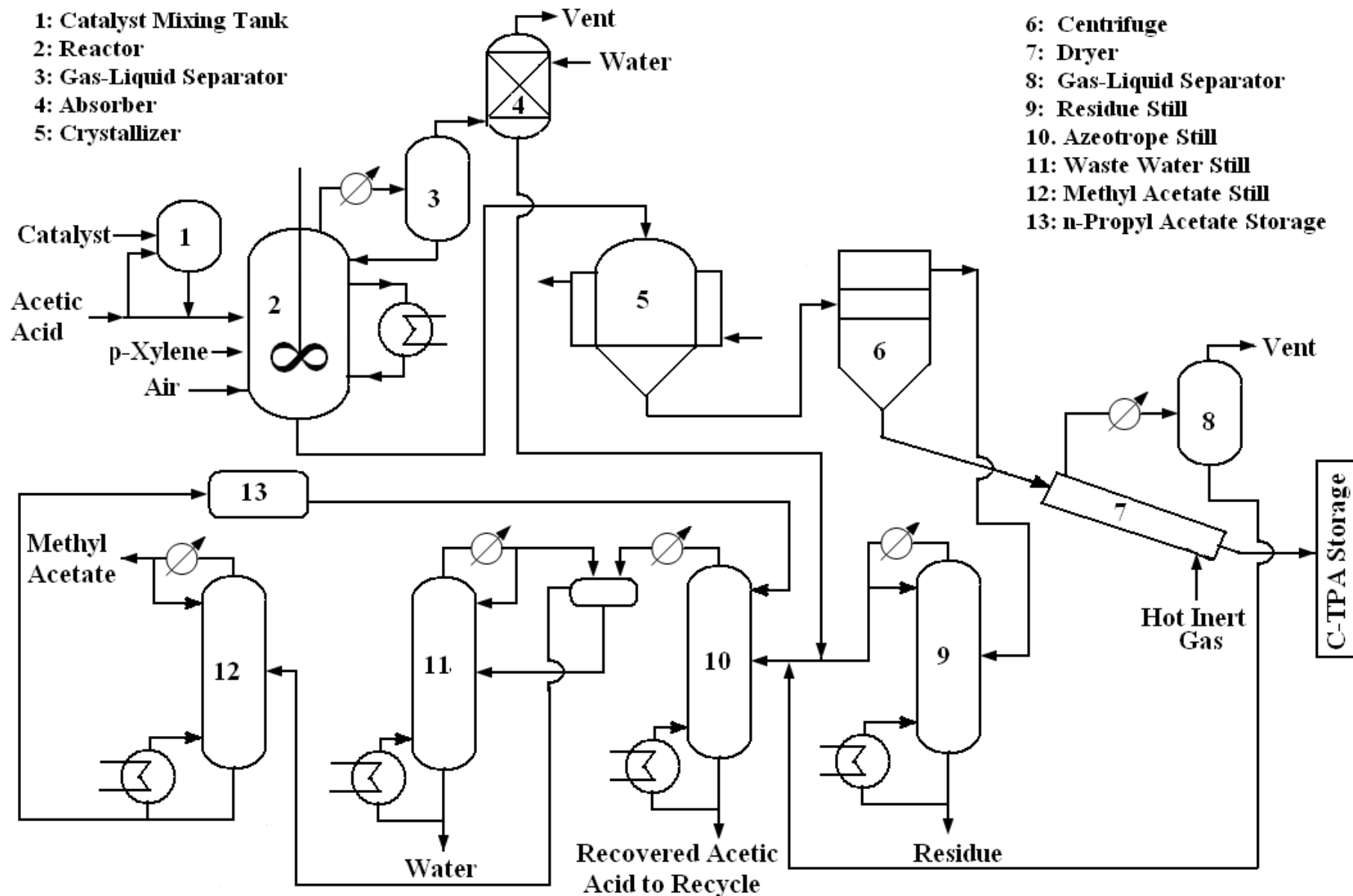




Catalyst: Mn or Co salt with Br as promoter
 Or Manganous and Cobaltous
 Bromide in acetic acid medium

- 1: Catalyst Mixing Tank
- 2: Reactor
- 3: Gas-Liquid Separator
- 4: Absorber
- 5: Crystallizer

- 6: Centrifuge
- 7: Dryer
- 8: Gas-Liquid Separator
- 9: Residue Still
- 10: Azeotrope Still
- 11: Waste Water Still
- 12: Methyl Acetate Still
- 13: n-Propyl Acetate Storage

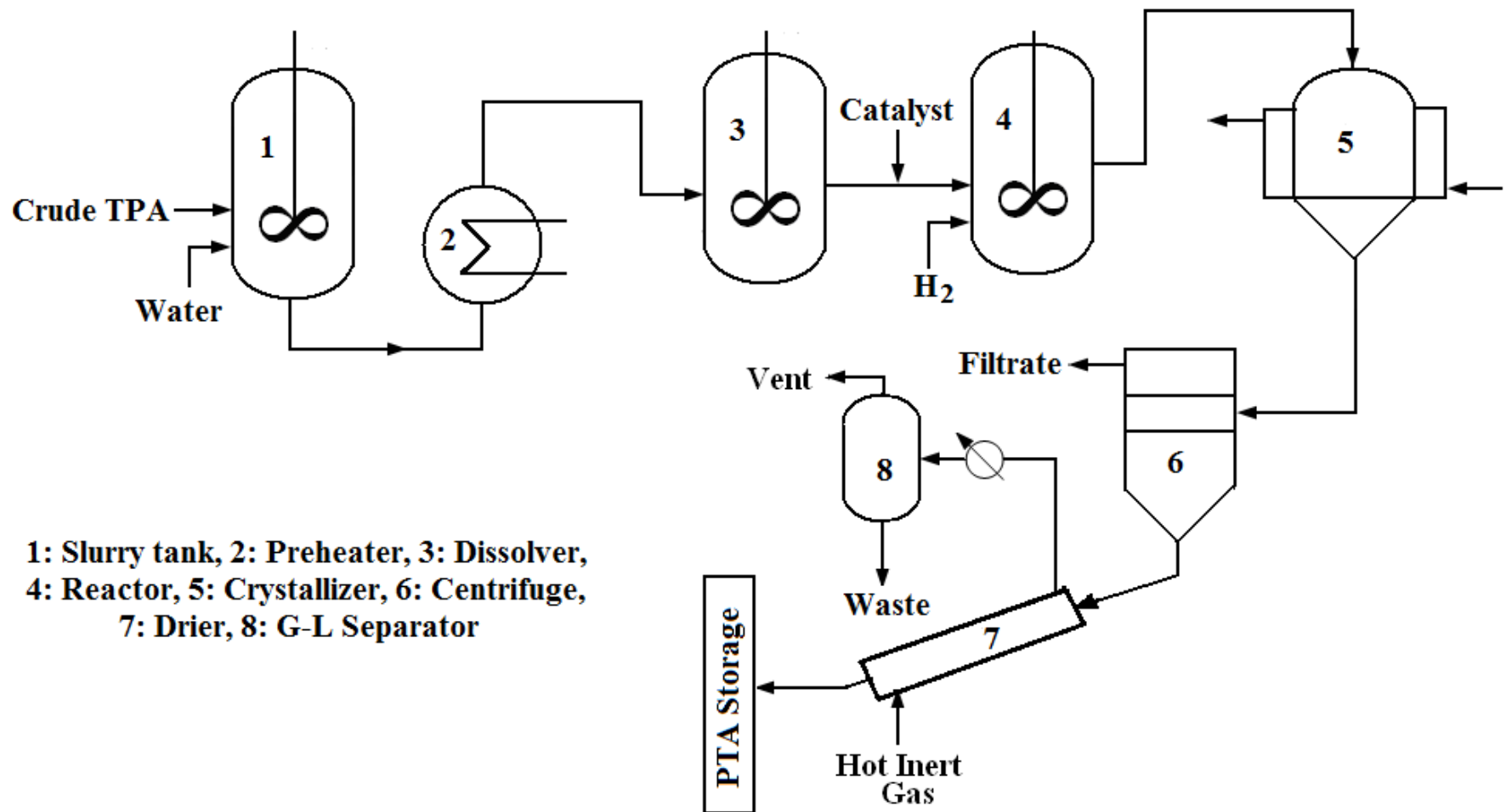
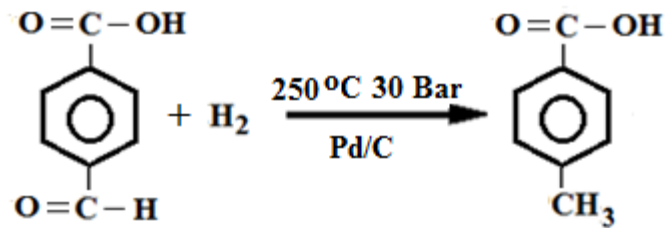


Production of Terephthalic Acid

Specification for Polymer Grade TPA

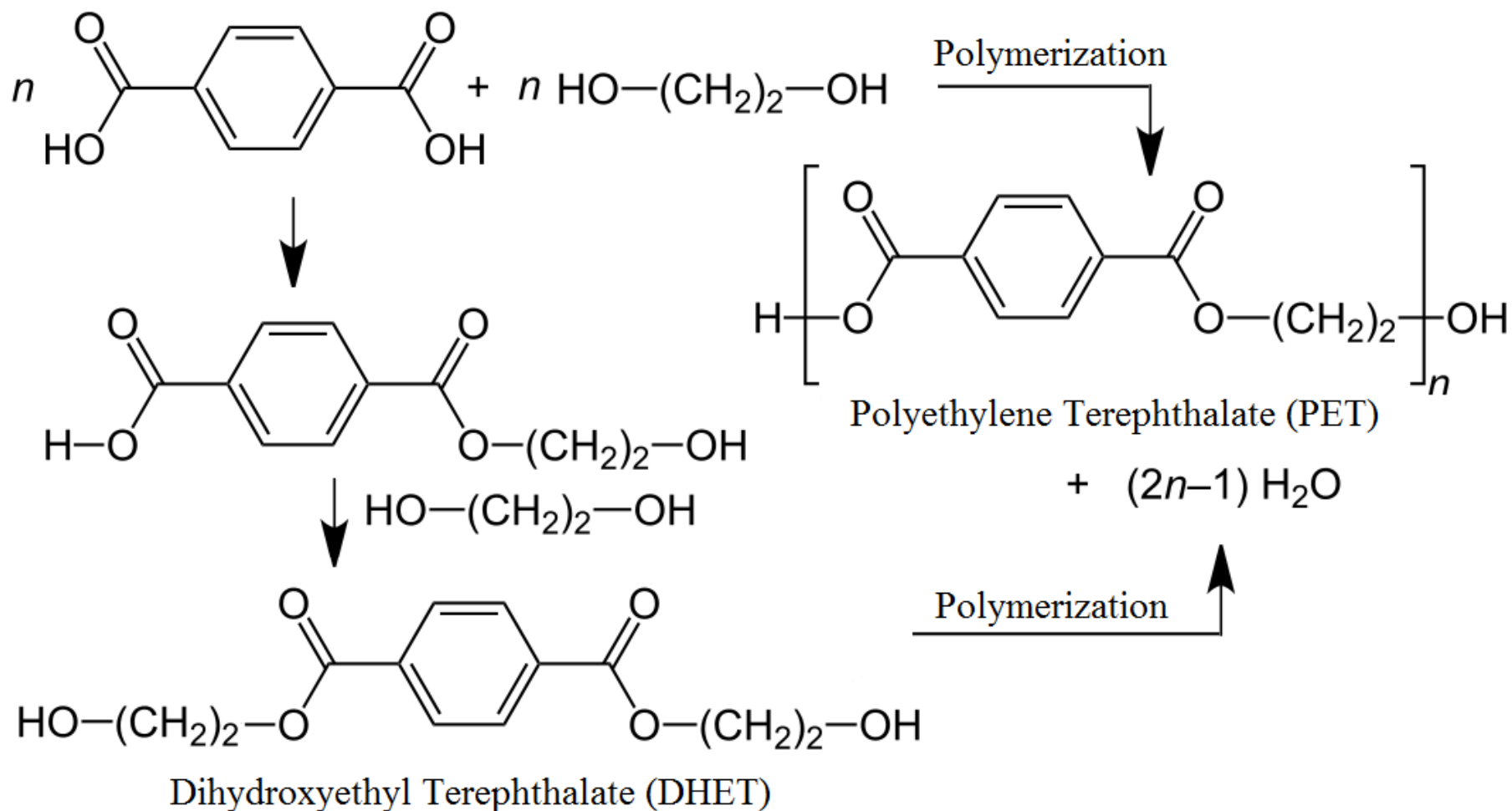
Property	Specification
Acid number, mg KOH/gm	675±2
Ash, ppm	15 max
Total significant metals, ppm (Mo, Cr, Ni, Co, Fe, Ti, Mg)	10 max
4-Formyl benzoic acid, ppm	25 max
Moisture, wt%	0.5 max
5% Dimethyl formamide solution colour, APHA*	10 max

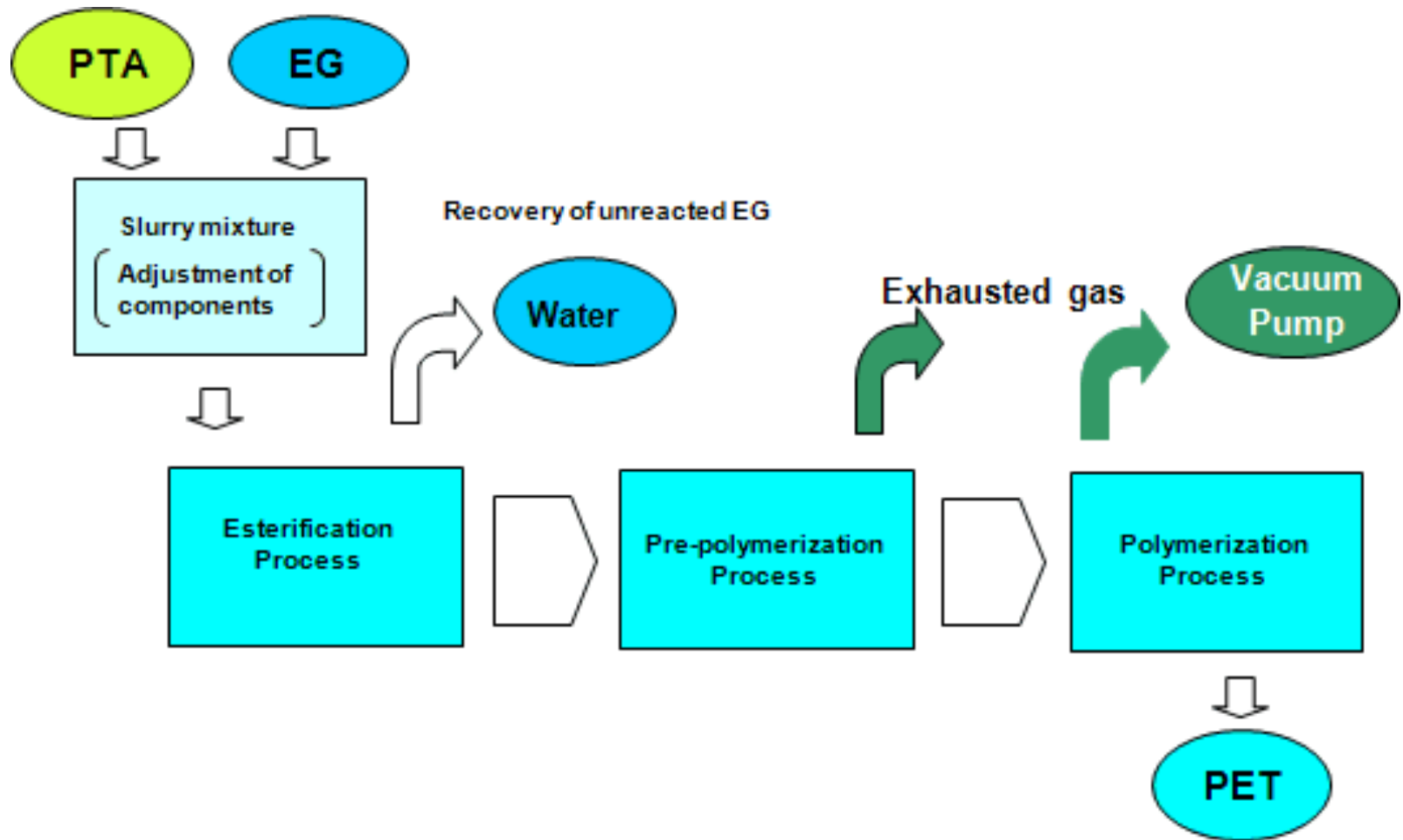
***APHA: American Public Health Association**



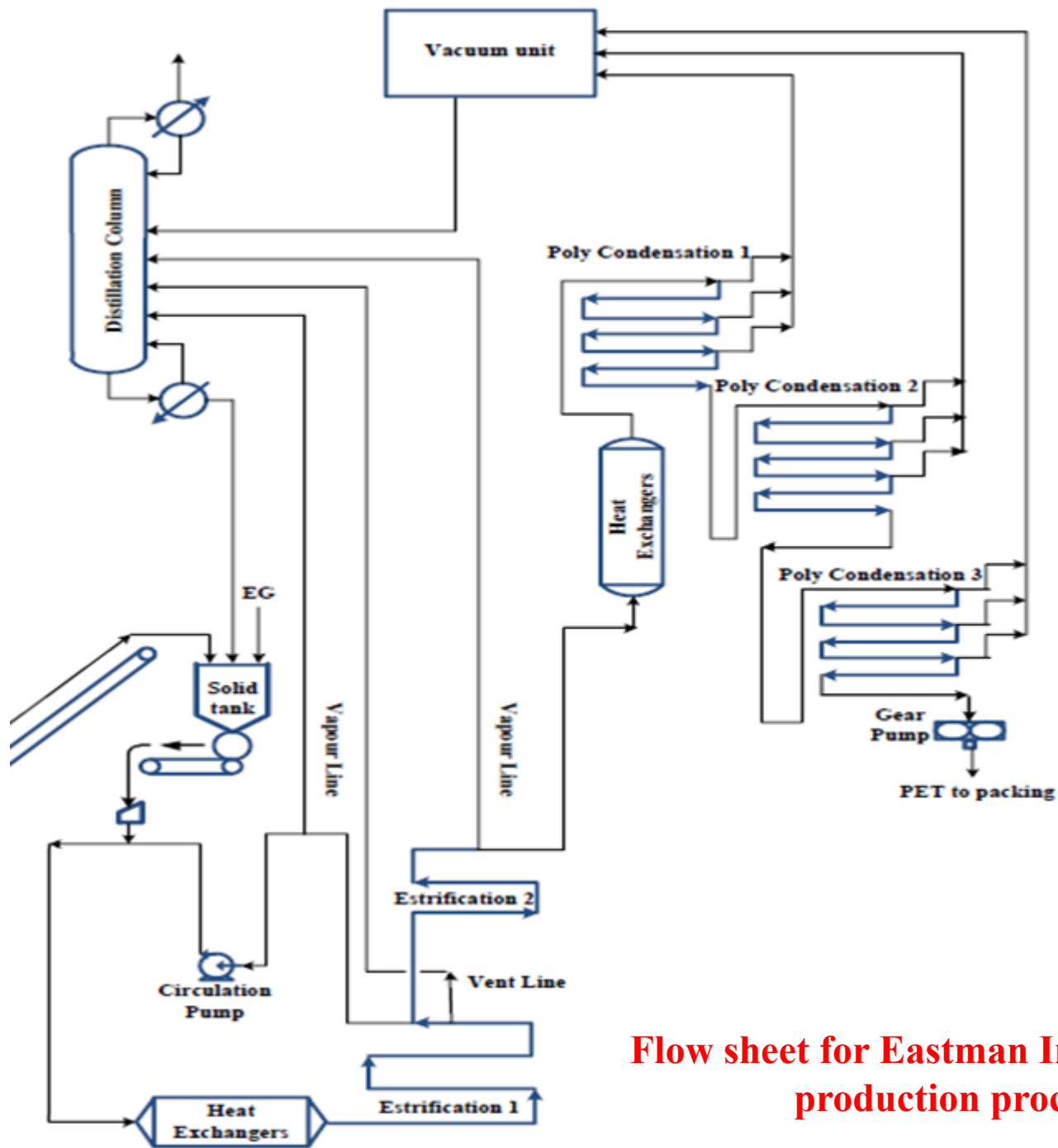
Production of Purified Terephthalic Acid (PTA)

Synthesis of Polyester, Polyethylene Terephthalate (PET)

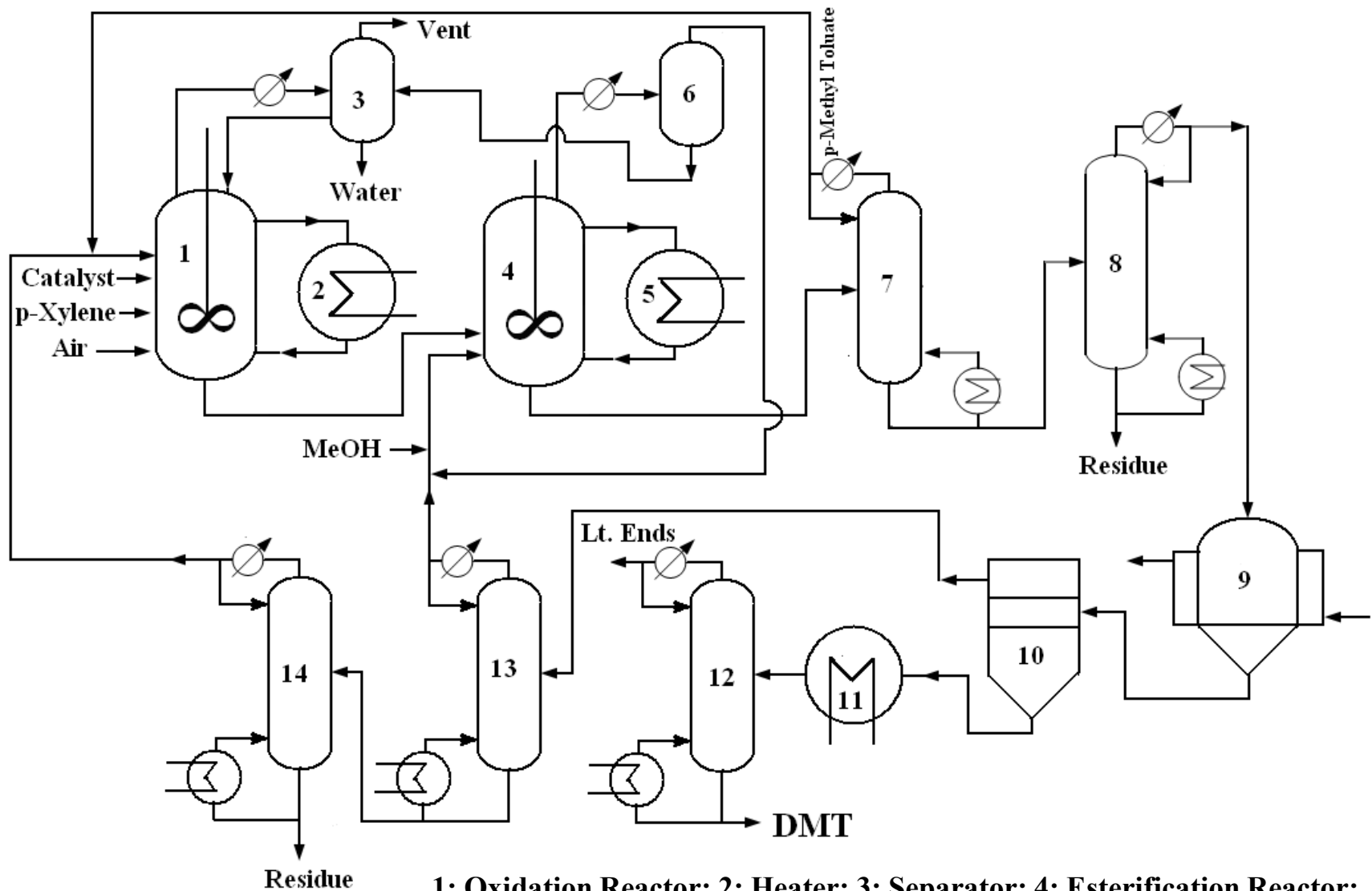




Block Diagram of PET Production Process

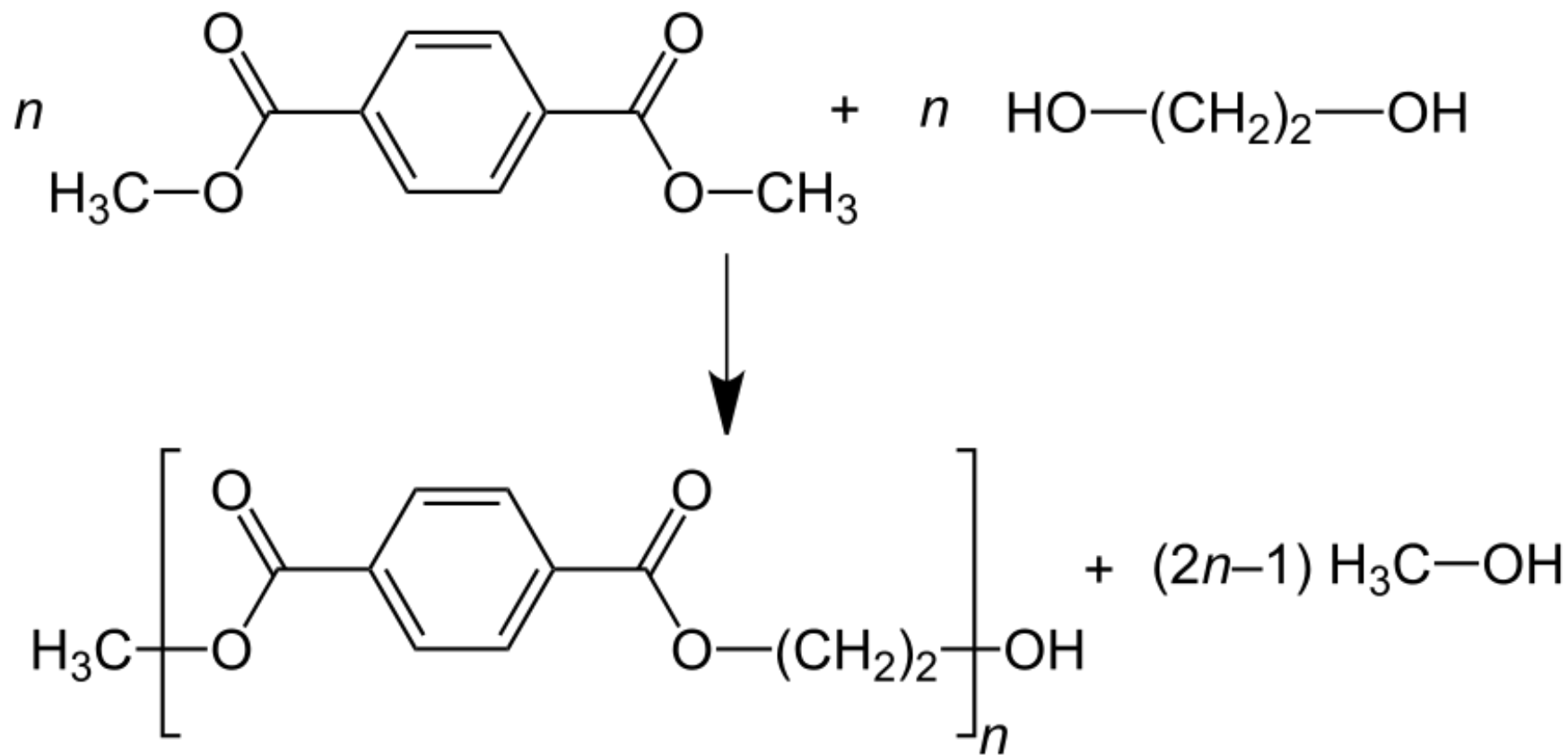


Flow sheet for Eastman IntegRex PET production process

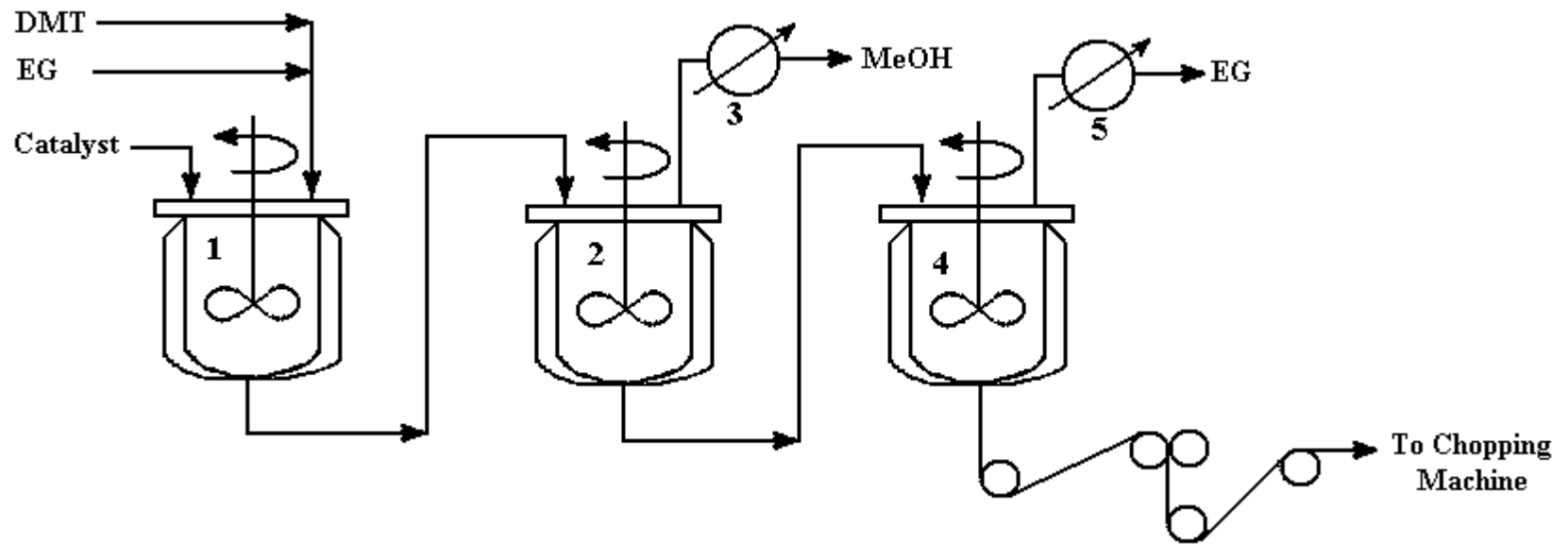


1: Oxidation Reactor; 2: Heater; 3: Separator; 4: Esterification Reactor;
5: Heater; 6: Reactants Recovery Column; 7: By-product Recovery Column;
8: Purification Column; 9: Crystallizer; 10: Centrifuge; 11: Heater; 12: DMT
Column; 13: MeOH Recovery Column; 14: Residue Still

Production of Dimethyl Terephthalate (DMT)



Polyethylene Terephthalate (PET)



Production of Polyethylene Terephthalate (PET)